

Authored by: Arpit Raj, LNMIIT Jaipur

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LeetCode 1696: Jump Game VI

Asked in AQR Capital and MathWorks

Approach 1: Initial DP Attempt

Explanation & Logic:

- since max score
- my first thought was to take only 1 step and keep going to end
- but array also has negative integers
- we can use dp , to store max score gained to reach this index
- basically ek index pe peeche se hi aa skte
- to apne range me best peeche waala dekho

```
class Solution {
public:
    int maxResult(vector<int>& nums, int k) {

        //since max score
        //my first thought was to take only 1 step and keep going to end
        //but array also has negative integers

        //we can use dp , to store max score gained to reach this index
        int n=nums.size();
        vector<int> dp(n,0);
        dp[0]=nums[0];

        for(int i=1;i<n;i++){

            int best=INT_MIN;
            for(int j=max(0,i-k);j<i;j++){
                //basically ek index pe peeche se hi aa skte
                //to apne range me best peeche waala dekho

                best=max(best,dp[j]);
                dp[i]=best+nums[i];
            }
        }
    }
}
```

```
        return dp[n-1];  
    }  
};
```

Approach 2: Optimized using Monotonic Queue

Explanation & Logic:

- earlier i tried normal dp , it was giving tle
- so to optimize
- so i thought of using a monnotonic queue
- queue ke front me max rahgea
- agar koi bhi usse bada aaya to smaller gain waalo ka koi kaam nhi rhega hata denge
- well store index kyunki out of range ko hatana h

```
class Solution {  
public:  
    int maxResult(vector<int>& nums, int k) {  
  
        //since max score  
        //my first thought was to take only 1 step and keep going to end  
        //but array also has negative integers  
  
        //we can use dp , to store max score gained to reach this index  
  
        //earlier i tried normal dp , it was giving tle  
        //so to optimize  
  
        //so i thought of using a monnotonic queue  
        //queue ke front me max rahgea  
        //agar koi bhi usse bada aaya to smaller gain waalo ka koi kaam nhi  
        rhenga hata denge  
  
        int n=nums.size();  
        vector<int> dp(n,0);  
        dp[0]=nums[0];  
  
        deque<int> dq;  
        dq.push_back(0); //well store index kyunki out of range ko hatana h  
  
        for(int j=1;j<n;j++){  
  
            while(!dq.empty() && dq.front()<j-k){  
                dq.pop_front();  
            }  
  
            dp[j]=nums[j]+dp[dq.front()];  
        }  
    }  
};
```

```
        while(!dq.empty() && dp[dq.back()]<dp[j]){
            dq.pop_back();
        }

        dq.push_back(j);

    }

    return dp[n-1];
};
```